

PHYSICS

1. A ball is thrown vertically upwards with a certain initial velocity. At half of its maximum height, its speed is v . What is the maximum height reached by the ball?

- A) v^2/g B) $2v^2/g$ C) $v^2/2g$ D) v/g

2. Two forces of equal magnitude act simultaneously on a particle. If the magnitude of their resultant force is equal to the magnitude of either of the individual forces, what is the angle between the two forces?

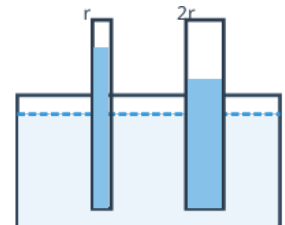
- A) 60° B) 90° C) 120° D) 180°

3. If the radius of the Earth suddenly shrinks by 1% while its mass remains constant, the acceleration due to gravity (g) on its surface will:

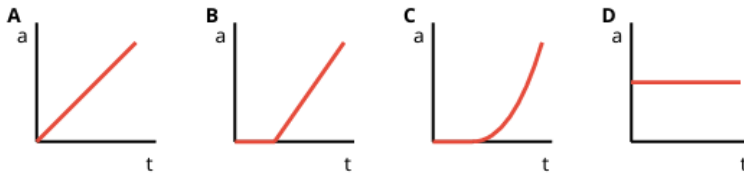
- A) Decrease by 1% B) Decrease by 2% C) Increase by 1% D) Increase by 2%

4. A capillary tube of radius r is immersed in water and the water rises to a height h . The mass of the water in the capillary is m . If another capillary tube of radius $2r$ is taken and dipped in water, what will be the mass of the water that rises in it?

- A) $m/2$ B) m
C) $2m$ D) $4m$



5. A block of mass m is initially at rest on a rough horizontal surface. A time-dependent horizontal force $F = kt$ (where k is a positive constant and t is time) is applied to the block. Which of the following statements correctly describes the motion graph?



- A) Graph A B) Graph B
C) Graph C D) Graph D

6. A uniform heavy chain of length L and mass M is lying on a smooth horizontal table, with one-third of its length hanging vertically over the edge. What is the minimum work required to pull the hanging part completely onto the table?

- A) $MgL/3$ B) $MgL/6$ C) $MgL/9$ D) $MgL/18$

7. A solid sphere and a thin hollow cylinder roll down an inclined plane without slipping. Both start from rest at the same height. Which one reaches the bottom first?

- A) The solid sphere B) The hollow cylinder C) Both reach simultaneously D) Depends on masses

8. A simple pendulum consisting of a string of length L and a bob of mass m is oscillating with an angular amplitude θ_0 . The tension in the string when the bob passes through the lowest point of its trajectory is:

- A) $mg(1 + 2 \cos \theta_0)$ B) $mg(3 - 2 \cos \theta_0)$ C) $mg(1 - \cos \theta_0)$ D) $mg(2 - \cos \theta_0)$

9. A smooth ball of mass m strikes a perfectly smooth horizontal floor at an angle of 45° with the vertical axis. If the coefficient of restitution of the collision is $e = 0.5$, what is the angle of reflection (with respect to the vertical)?

- A) $\tan^{-1}(0.5)$ B) $\tan^{-1}(1)$ C) $\tan^{-1}(2)$ D) $\tan^{-1}(0.25)$

10. Two mass blocks m_1 and m_2 ($m_1 > m_2$) are connected by a massless inextensible string passing over a solid cylindrical pulley of mass M and radius R . The string does not slip on the pulley. What is the downward acceleration of the mass m_1 ?

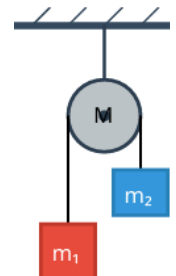
- A) $(m_1 - m_2)g/(m_1 + m_2)$ B) $(m_1 - m_2)g/(m_1 + m_2 + M)$
C) $(m_1 - m_2)g/(m_1 + m_2 + M/2)$ D) $m_1 g/(m_1 + m_2 + 2M)$

11. A faulty thermometer reads 5°C in melting ice and 95°C in steam at standard atmospheric pressure. What is the correct temperature when this thermometer reads 41°C ?

- A) 40°C B) 41°C C) 36°C D) 45°C

12. If the root mean square (rms) speed of ideal gas molecules in a closed rigid container is halved, what happens to the pressure of the gas?

- A) It becomes half B) It becomes one-fourth C) It remains the same D) It doubles



13. A uniform metal sheet with a circular hole is heated. As the temperature increases, the diameter of the hole will:

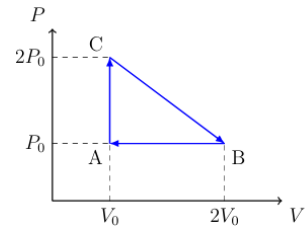
- A) Decrease B) Increase C) Remain the same D) First decrease, then increase

14. During an isothermal expansion of an ideal gas, the work done by the gas is strictly equal to:

- A) The change in internal energy B) The heat absorbed by the gas C) Zero D) The change in enthalpy

15. Consider the cyclic process ACBA for an idea gas as shown in the P-V diagram. What is the total net work done by the gas in one complete cycle?

- A) $\frac{1}{2}P_0V_0$ B) $-\frac{1}{2}P_0V_0$
C) P_0V_0 D) $-P_0V_0$



16. A Carnot engine operates between a heat source at T_1 and a heat sink at T_2 ($T_1 > T_2$). If the temperature of the source is increased by ΔT , the efficiency increases by $\Delta\eta_1$.

Alternatively, if the temperature of the sink is decreased by the same ΔT , the efficiency increases by $\Delta\eta_2$. Which relation is correct?

- A) $\Delta\eta_1 = \Delta\eta_2$ B) $\Delta\eta_1 > \Delta\eta_2$ C) $\Delta\eta_1 < \Delta\eta_2$ D) Depends strictly on T_1 and T_2

17. 10 g of ice at 0°C is mixed with 100 g of water at 50°C in a calorimeter of negligible heat capacity. What is the resultant temperature of the mixture?

- A) 38.2°C B) 31.2°C C) 40.0°C D) 45.5°C

18. Two sound waves propagate through two different diatomic gases, Hydrogen (H_2) and Oxygen (O_2), maintained at identical temperature and pressure. If the pressure amplitude of both sound waves is exactly the same, what is the ratio of the intensity of the sound wave in Hydrogen to that in Oxygen ($I_{\text{H}_2} : I_{\text{O}_2}$)?

- A) 1:4 B) 1:16 C) 4:1 D) 16:1

19. In a single-slit diffraction experiment, light of wavelength λ illuminates a slit of width a . For what precise value of the slit width a will the first minimum fall at a diffraction angle of exactly 30° ?

- A) $a = \lambda/2$ B) $a = \lambda$ C) $a = 2\lambda$ D) $a = 4\lambda$

20. Imagine a thin plano-convex lens made of glass ($\mu = 1.5$) with a focal length f . If the flat (plane) surface of this lens is silvered, the system acts as a concave mirror with focal length F_1 . If the curved surface is silvered instead, it acts as a concave mirror with focal length F_2 . What is the ratio F_1/F_2 ?

- A) 1:1 B) 1:2 C) 2:1 D) 3:1

21. A sonometer wire vibrates in its fundamental mode at a frequency f when the tension is provided by a solid metallic mass M hanging freely in air. When completely submerged in water, the fundamental frequency drops to $f/\sqrt{2}$. What is the specific gravity of the metallic mass?

- A) 1.5 B) 2.0 C) 2.5 D) $\sqrt{2}$

22. In a YDSE apparatus, white light is used to illuminate two slits separated by a distance d . A screen is placed at a distance D ($D \gg d$). Consider a specific point P on the screen that lies exactly straight across in front of the upper slit S_1 . Which wavelength will be completely missing at point P ?

- A) $\lambda = d^2/D$ B) $\lambda = d^2/2D$ C) $\lambda = 2d^2/D$ D) $\lambda = d^2/4D$

23. A whistle emitting a continuous sound of frequency 400 Hz is tied to the end of a string 1 m long and rotated in a horizontal circle with a constant angular velocity of 20 rad/s. An observer is standing at a large distance away in the exact same plane as the circle. If the speed of sound is 340 m/s, what is the minimum frequency recorded?

- A) 377.8 Hz B) 380.0 Hz C) 400.0 Hz D) 425.0 Hz

24. A monochromatic ray of light travels through an equilateral glass prism ($A = 60^\circ$). The ray undergoes minimum deviation, and the angle of incidence is exactly equal to $3/4$ of the angle of the prism. What is the precise speed of light inside the prism? ($c = 3 \times 10^8$ m/s)

- A) 1.5×10^8 m/s B) 2.12×10^8 m/s C) 2.25×10^8 m/s D) 2.8×10^8 m/s

25. To perfectly resolve two closely spaced spectral lines of sodium lamp (600.0 nm and 600.5 nm) in the third order of the diffraction spectrum, what is the absolute minimum number of ruled lines on the grating that the light beam must illuminate?

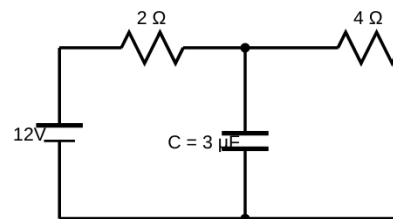
- A) 200 B) 400 C) 800 D) 1200

26. According to Gauss's law, what is the total electric flux passing through a closed 3D Gaussian surface that completely encloses a single electric dipole?

- A) q/ϵ_0 B) $2q/\epsilon_0$ C) Zero D) Infinity

27. In the DC circuit shown below, the system has reached a steady state. Calculate the steady-state electric charge stored on the $3\ \mu\text{F}$ capacitor.

- A) $12\ \mu\text{C}$ B) $24\ \mu\text{C}$
C) $36\ \mu\text{C}$ D) Zero



28. A neutral soap bubble is gently given a strictly negative electrostatic charge.

Due to this process, the overall radius of the soap bubble will:

- A) Decrease significantly B) Remain exactly the same C) Increase D) First decrease, then increase

29. An ideal parallel plate capacitor with initial capacitance C_0 is fully charged to a potential difference V_0 by a battery. The battery is then completely disconnected. An external agent pulls the plates apart until the distance between them is exactly doubled. The mechanical work done by the external agent is:

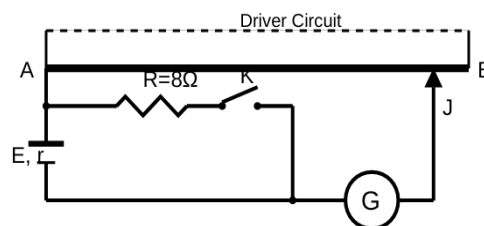
- A) $1/2 C_0 V_0^2$ B) $C_0 V_0^2$ C) $-1/2 C_0 V_0^2$ D) Zero

30. In the microscopic vector form of Ohm's Law, the current density J is related to the electric field E and the conductivity σ of the material by the equation:

- A) $J = \sigma/E$ B) $J = E/\sigma$ C) $J = \sigma E$ D) $J = \sigma E^2$

31. In the potentiometer experiment diagrammed below, a primary cell of EMF E is balanced at length $l_1 = 100\text{ cm}$ when the switch K is open. When switch K is closed, connecting the resistance box $R = 8\ \Omega$ in parallel with the cell, the balance point shifts to $l_2 = 80\text{ cm}$. What is the internal resistance r of the cell E ?

- A) $1.0\ \Omega$ B) $1.6\ \Omega$
C) $2.0\ \Omega$ D) $4.0\ \Omega$



32. For a given thermocouple, what is the value of the thermoelectric power (Seebeck coefficient, $S = dE/dT$) specifically at the neutral temperature?

- A) Maximum positive value B) Maximum negative value C) Constant non-zero value D) Zero

33. In the study of ferromagnetic materials, the total area enclosed within the B - H hysteresis loop strictly represents the:

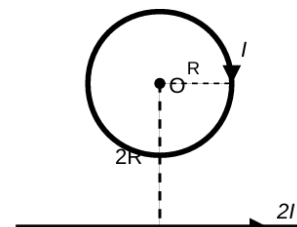
- A) Magnetic susceptibility of the material. B) Retentivity of the material.
C) Energy dissipated as heat per unit volume per cycle. D) Total magnetic flux density.

34. An electron enters a uniform magnetic field and follows a perfectly circular orbit. The total work done by the magnetic Lorentz force on the electron during one complete revolution is:

- A) $\mu_0 e v B$ B) $1/2 mv^2$ C) Zero
D) $2\pi r e B v$

35. A circular coil of radius R carries a current I . Coplanar to it is an infinitely long straight wire carrying a current $2I$, located at a perpendicular distance of $2R$ from the center of the coil (as shown). What is the ratio of the magnitude of the magnetic field at the center of the coil due to the coil, to the magnetic field at the center due to the straight wire ($B_{\text{coil}} : B_{\text{wire}}$)?

- A) $\pi : 1$ B) $1 : \pi$ C) $2\pi : 1$ D) $1 : 2$



36. In a series LRC alternating current circuit, when the circuit is exactly at resonance, the total impedance of the circuit becomes:

- A) Zero B) Maximum and purely reactive C) Minimum and purely resistive D) Infinite

37. A transformer and a full-wave diode bridge rectifier are used to convert mains AC into DC. If the frequency of the input alternating voltage is 50 Hz, what is the fundamental frequency of the ripples in the output pulsating DC voltage?

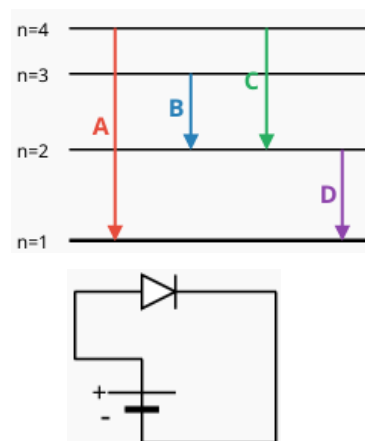
- A) 25 Hz B) 50 Hz C) 100 Hz D) 0 Hz

38. Two ideal inductors with self-inductances L_1 and L_2 are connected in series. If they are perfectly coupled magnetically (coefficient of coupling $K = 1$) such that their magnetic fluxes completely aid each other, the equivalent inductance of the combination is:

- A) $L_1 + L_2$ B) $(\sqrt{L_1} + \sqrt{L_2})^2$ C) $|L_1 - L_2|$ D) $(L_1 L_2) / (L_1 + L_2)$

39. Which electronic transition in a hydrogen atom corresponds to the emission of a photon with the longest wavelength in the visible region of the hydrogen spectrum?

- A) Transition A B) Transition B
C) Transition C D) Transition D



40. Consider a semiconductor circuit diagram where the P-side of a P-N junction diode is connected to the positive terminal of a DC battery and the N-side to the negative terminal. In this configuration, what happens to the internal properties?

- A) The depletion layer widens, increasing resistance.
B) The depletion layer narrows, allowing majority charge carriers to cross easily.
C) The diode enters a state of Zener breakdown.
D) Only minority charge carriers contribute to current.

41. In the Bohr model of the hydrogen atom, what is the ratio of the shortest wavelength in the Balmer series to the shortest wavelength in the Lyman series?

- A) 1:4 B) 4:1 C) 9:1 D) 27:5

42. The binding energy per nucleon for Deuterium (^2H) is 1.1 MeV and for Helium (^4He) is 7.1 MeV. If two Deuterium nuclei fuse to form a single Helium nucleus, what is the approximate energy released?

- A) 24.0 MeV B) 28.4 MeV C) 26.2 MeV D) 12.0 MeV

43. An alpha particle and a proton are accelerated from rest through the exact same potential difference V . What is the ratio of their final de-Broglie wavelengths ($\lambda_\alpha : \lambda_p$)?

- A) 1:2 B) $1:2\sqrt{2}$ C) $2\sqrt{2}:1$ D) 1:4

44. A radioactive sample's activity drops to 1/16 of its original initial value in 120 days. What is the half-life of this radioactive substance?

- A) 15 days B) 20 days C) 30 days D) 60 days

45. In a digital logic circuit, the output of a standard NAND gate is directly connected to both inputs of a standard NOR gate. The overall equivalent logic behavior of this combination is identical to that of an:

- A) AND gate B) OR gate C) XOR gate D) XNOR gate

46. According to Hubble's law regarding the expansion of the universe, the recession velocity of a distant galaxy is directly proportional to its distance from the observer. What is the standard SI unit of the Hubble constant (H_0)?

- A) $\text{m} \cdot \text{s}^{-1}$ B) s^{-1} C) $\text{m} \cdot \text{s}^{-2}$ D) Light-years per second

47. In an X-ray tube, electrons are accelerated by a potential difference of 50 kV. What is the minimum wavelength (cut-off wavelength) of the continuous X-ray spectrum produced? (Assume $hc = 1240 \text{ eV} \cdot \text{nm}$).

- A) 0.0124 nm B) 0.0248 nm C) 0.248 nm D) 2.48 nm

48. In Millikan's oil drop experiment, an oil drop carrying a negative charge is held stationary between two horizontal charged plates. If the electric field is suddenly switched off, the oil drop will immediately begin to:

- A) Fall with a constant velocity equal to its terminal velocity.
B) Fall with a constant acceleration of g forever.
C) Fall with an initial acceleration, which eventually decreases to zero as terminal velocity is reached.
D) Move upwards due to buoyant forces.

49. In the Standard Model of particle physics, all elementary particles are classified by quantum numbers. What is the assigned baryon number of an electron?

- A) +1 B) -1 C) +1/2 D) 0

50. A Zener diode is a specially designed heavily doped semiconductor device. To utilize its primary property of acting as a voltage regulator, it must be operated in which configuration?

- A) Forward bias region B) Reverse breakdown region C) Zero bias equilibrium state D) Saturation current region

Chemistry

51. A metal M forms two oxides. The higher oxide contains 80% metal. If 0.72 g of the lower oxide gave 0.8 g of the higher oxide when oxidized, what is the equivalent weight of the metal in the higher oxide?

- A) 8 B) 16 C) 32 D) 64

52. A mixture of 20 mL of CO, CH₄ and N₂ is exploded with 10 mL of O₂. After cooling, the volume is 14 mL. On treatment with KOH, the volume reduces to 8 mL. What is the volume of CH₄ in the original mixture? (Assume all measurements at same T and P).

- A) 2 mL B) 4 mL C) 6 mL D) 8 mL

53. If the kinetic energy of a free electron is doubled, its de-Broglie wavelength changes by a factor of:

- A) 1/2 B) $\sqrt{2}$ C) $1/\sqrt{2}$ D) 2

54. Which of the following processes requires the largest amount of energy?

- A) $\text{Al(g)} \rightarrow \text{Al}^+(\text{g}) + \text{e}^-$ B) $\text{Al}^+(\text{g}) \rightarrow \text{Al}^{2+}(\text{g}) + \text{e}^-$ C) $\text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g}) + \text{e}^-$ D) $\text{Al}^{3+}(\text{g}) \rightarrow \text{Al}^{4+}(\text{g}) + \text{e}^-$

55. The dipole moment of H₂O is 1.85 D, and the H-O-H bond angle is 104.5°. Which of the following is the primary reason the dipole moment of H₂S is significantly lower (0.95 D) despite having the same VSEPR geometry?

- A) The S-H bond is longer than the O-H bond. B) Sulfur is much less electronegative than Oxygen. C) The H-S-H bond angle is closer to 90° reducing vector addition. D) Both B and C contribute significantly.

56. In the disproportionation reaction of Phosphorus (P₄) in alkaline medium to form Phosphine (PH₃) and Hypophosphite ion (H₂PO₂⁻), what is the equivalent weight of P₄? (M = molar mass of P₄)

- A) 3M/4 B) M/12 C) 3M/8 D) M/3

57. The ratio of the root mean square velocity of H₂ at 50 K to that of O₂ at 800 K is:

- A) 1:1 B) 1:4 C) 4:1 D) 1:2

58. For the reaction $\text{A(s)} \rightleftharpoons \text{B(g)} + \text{C(g)}$, the equilibrium constant K_p is 0.04 atm² at a certain temperature. If a mixture containing only B and C initially at a partial pressure of 0.3 atm each is allowed to reach equilibrium, what will be the final total pressure?

- A) 0.6 atm B) 0.4 atm C) 0.2 atm D) 0.8 atm

59. A 100 mL solution of 0.1 N Na₂CO₃ is titrated with 0.1 N HCl using phenolphthalein as an indicator. What volume of HCl is required to reach the end point?

- A) 100 mL B) 50 mL C) 200 mL D) 25 mL

60. What is the pH of a 10⁻⁸ M solution of NaOH at 25°C? (Given log 1.05 ≈ 0.02)

- A) 8.00 B) 6.00 C) 7.04 D) 6.96

61. The K_{sp} of Ag₂S is 10⁻⁵¹. In a solution saturated with H₂S (0.1 M) and buffered at pH = 3, what is the maximum concentration of Ag⁺ that can remain in solution without precipitating? (Given for H₂S: K_{a1} * K_{a2} = 10⁻²¹)

- A) 10⁻¹⁷ M B) 10⁻¹⁸ M C) 10⁻²⁴ M D) 10⁻³³ M

62. In a zero-order reaction, if the initial concentration of the reactant is doubled, the half-life period will:

- A) Remain constant B) Be doubled C) Be halved D) Increase four times

63. The activation energy for the forward reaction is 50 kJ/mol, and the enthalpy change (ΔH) for the reaction is -20 kJ/mol. What is the activation energy for the reverse reaction?

- A) 30 kJ/mol B) 70 kJ/mol C) 50 kJ/mol D) 20 kJ/mol

64. During the electrolysis of an aqueous solution of CuSO₄ using platinum electrodes, exactly 2 Faradays of electricity are passed. What is the mass of copper deposited at the cathode and the volume of gas evolved at the anode at STP? (Atomic mass of Cu = 63.5 g/mol)

- A) 63.5 g Cu and 11.2 L O₂ B) 31.75 g Cu and 22.4 L O₂ C) 63.5 g Cu and 22.4 L H₂ D) 127 g Cu and 11.2 L O₂

65. For the isothermal reversible expansion of 1 mole of an ideal gas from volume V₁ to V₂ at temperature T, the work done (W) by the system is given by:

- A) $W = -RT \ln(V_2/V_1)$ B) $W = RT \ln(V_2/V_1)$ C) $W = -P_{\text{ext}}(V_2 - V_1)$ D) $W = nC_v \Delta T$

66. When water freezes into ice at 0°C and 1 atm pressure, the entropy change of the system (ΔS_{sys}) and the total entropy change of the universe (ΔS_{univ}) are respectively:

- A) Positive, Positive B) Negative, Zero C) Negative, Positive D) Negative, Negative

67. In the uranium radioactive decay series, $^{238}\text{U}_{92}$ decays to the stable isotope $^{206}\text{Pb}_{82}$. How many alpha (α) and beta (β) particles are emitted during this entire sequence?

- A) 6 α , 8 β B) 8 α , 6 β C) 8 α , 8 β D) 6 α , 6 β

68. When zinc reacts with dilute sulfuric acid, it readily reduces ferric chloride to ferrous chloride. However, bubbling ordinary molecular hydrogen through ferric chloride produces no reaction. This indicates that nascent hydrogen is:

- A) A more powerful oxidizing agent than molecular hydrogen.
B) Associated with excess chemical energy at the moment of its generation, making it highly reactive.
C) Isotopic with deuterium, giving it higher kinetic energy.
D) A catalyst that lowers the activation energy of ferric chloride.

69. In the aluminothermite process (Thermite welding) used for joining broken railway tracks, what is the specific chemical role of the aluminum powder?

- A) It acts as a basic flux to remove silica impurities.
B) It serves as a strong reducing agent due to its high affinity for oxygen, reducing iron(III) oxide to molten iron.
C) It lowers the melting point of the iron oxide to facilitate smelting.
D) It alloys with iron to form high-strength steel.

70. Chronic occupational or environmental exposure to which heavy metal, specifically listed in the bio-inorganic syllabus, leads to 'plumbism', a condition characterized by severe anemia and a distinctive blue line along the gums?

- A) Cadmium (Cd) B) Mercury (Hg) C) Arsenic (As) D) Lead (Pb)

71. Heavy water (D_2O) is extensively used as a moderator in nuclear power reactors. What is its precise physical function in this process?

- A) To absorb excess neutrons and stop the chain reaction.
B) To slow down fast-moving fission neutrons so they can effectively be captured by Uranium-235.
C) To cool down the reactor core by absorbing heat generated by fission.
D) To initiate the fission process by bombarding the uranium nucleus.

72. While sodium carbonate (Na_2CO_3) is successfully manufactured on a large scale using the Solvay process, potassium carbonate (K_2CO_3) cannot be prepared by this same method. Why?

- A) Potassium chloride is insoluble in the ammoniacal brine solution.
B) Potassium bicarbonate (KHCO_3) is highly soluble in water and does not precipitate out.
C) Potassium reacts explosively with ammonia gas.
D) The lattice energy of potassium carbonate is too high to be formed.

73. Zone refining is an electro-metallurgical technique employed to obtain metals of ultra-high purity, such as silicon or germanium for semiconductors. What fundamental principle makes this technique work?

- A) Impurities vaporize at lower temperatures than the pure metal.
B) The pure metal has a higher density and settles at the bottom of the molten zone.
C) Impurities are more soluble in the molten state of the metal than in the solid state.
D) The pure metal is attracted to a strong magnetic field while impurities are not.

74. Despite being in the same group, ammonia (NH_3) exists as a liquid at much higher temperatures compared to phosphine (PH_3), which is a gas. What primary chemical phenomenon explains this physical anomaly?

- A) Phosphorus has a larger atomic radius than nitrogen.
B) Ammonia experiences strong intermolecular hydrogen bonding, whereas phosphine does not.
C) Phosphine forms coordinate covalent bonds with itself.
D) The N-H bond has a lower bond dissociation energy than the P-H bond.

75. When excess aqueous ammonia is added to a solution of blue vitriol ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), the solution turns a deep, inky blue. According to coordinate chemistry, what is the formula and shape of the resulting complex ion?

- A) $[\text{Cu}(\text{NH}_3)_6]^{2+}$, Octahedral B) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, Tetrahedral C) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, Square planar D) $[\text{Cu}(\text{OH})_4]^{2-}$, Square planar

76. During the extraction of copper from copper pyrites (CuFeS_2), silica (SiO_2) is intentionally added to the reverberatory furnace during the smelting process. What is its exact metallurgical function?

- A) It reduces the copper(I) oxide to metallic copper.
B) It acts as an acidic flux to chemically combine with the basic iron(II) oxide impurity, forming a removable iron silicate slag.
C) It prevents the oxidation of copper sulfide to copper oxide.
D) It increases the conductivity of the molten matte for electro-refinement.

77. When metallic copper reacts with dilute nitric acid (HNO_3), it does not produce hydrogen gas. Instead, it yields a colorless gas that immediately turns reddish-brown upon contact with air. Identify the two gases involved in this sequence.

- A) Nitrous oxide (N_2O) turning into Nitrogen dioxide (NO_2)
- B) Nitric oxide (NO) turning into Nitrogen dioxide (NO_2)
- C) Nitrogen dioxide (NO_2) turning into Dinitrogen tetroxide (N_2O_4)
- D) Nitrogen gas (N_2) turning into Nitric oxide (NO)

78. During the homolytic fission of a covalent bond, which of the following reactive intermediates is generated?

- A) Carbocation
- B) Carbanion
- C) Free radical
- D) Electrophile

79. Which of the following industrial processes is used for the manufacture of primary alcohols from alkenes using carbon monoxide and hydrogen?

- A) Hydroboration-oxidation
- B) Oxo-process
- C) Williamson's synthesis
- D) Fermentation

80. In Hoffmann's method for the separation of primary, secondary, and tertiary amines, which reagent is utilized?

- A) Hinsberg's reagent (Benzenesulfonyl chloride)
- B) Nitrous acid
- C) Diethyl oxalate
- D) Chloroform and alcoholic KOH

81. Rectified spirit, a specific type of ethanol mixture, consists of approximately:

- A) 100% pure ethanol
- B) 95.5% ethanol and 4.5% water
- C) Ethanol mixed with methanol and pyridine
- D) Ethanol containing 10% water

82. Williamson's synthesis of ethers generally follows which type of reaction mechanism?

- A) SN_1
- B) SN_2
- C) Electrophilic substitution
- D) Free radical substitution

83. What is the general formula representing a Grignard reagent (an organomagnesium compound)?

- A) R-Li
- B) R_2CuLi
- C) R-Mg-X
- D) R_2Cd

84. Which term describes the permanent displacement of sigma electrons along a saturated carbon chain due to the presence of an electron-withdrawing or electron-donating group?

- A) Resonance effect
- B) Inductive effect
- C) Electromeric effect
- D) Hyperconjugation

85. An SN_1 reaction of an optically active alkyl halide typically results in:

- A) Complete retention of configuration
- B) Complete inversion of configuration
- C) Racemization
- D) Formation of a meso compound

86. Which of the following compounds is an example of an organocopper compound mentioned in your syllabus?

- A) Alkylmagnesium halide
- B) Alkylolithium
- C) Dialkylcadmium
- D) Lithium dialkylcuprate (Gilman reagent)

87. Why is chlorobenzene significantly less reactive towards nucleophilic substitution reactions compared to chloroethane?

- A) The carbon atom in chlorobenzene is sp^3 hybridized, holding the chlorine atom tightly.
- B) The C-Cl bond acquires partial double bond character due to resonance.
- C) The benzene ring acts as a strong nucleophile, repelling other nucleophiles.
- D) Chlorobenzene is completely insoluble in organic solvents.

88. Aliphatic aldehydes are generally more reactive towards nucleophilic addition reactions than aliphatic ketones. What is the primary reason for this?

- A) Ketones have more steric hindrance and stronger electron-donating alkyl groups than aldehydes.
- B) Aldehydes have a higher boiling point than ketones.
- C) Ketones form more stable hydrates than aldehydes.
- D) The carbonyl carbon in aldehydes is less electrophilic than in ketones.

89. Why are carboxylic acids stronger acids than phenols and alcohols?

- A) Carboxylic acids contain two oxygen atoms, increasing hydrogen bonding.
- B) The carboxylate ion is highly stabilized by resonance involving two electronegative oxygen atoms.
- C) The phenoxide ion exhibits no resonance stabilization.
- D) The alkyl group in alcohols pushes electrons away, stabilizing the alkoxide ion.

90. Aniline is less basic than aliphatic primary amines (like methylamine). This is primarily because:

- A) Aniline is a larger molecule and sterically hindered.
- B) The lone pair of electrons on the nitrogen atom is delocalized into the benzene ring via resonance.
- C) The benzene ring has a +I (inductive) effect.
- D) Aniline forms stronger intermolecular hydrogen bonds.

91. Terminal alkynes (e.g., propyne) exhibit acidic character, whereas alkanes and alkenes generally do not. This acidity is due to:

- A) The presence of pi bonds which attract incoming bases.
- B) The high electronegativity of the sp hybridized carbon atom bonded to the hydrogen.
- C) The fact that terminal alkynes readily form free radicals.
- D) The stability of the resulting carbanion due to hyperconjugation.

92. What is the major product formed when propene undergoes hydroboration-oxidation (B_2H_6/THF followed by H_2O_2/OH^-)?

- A) Propan-2-ol
- B) Propan-1-ol
- C) Propane
- D) Propanal

93. When methylmagnesium bromide (CH_3MgBr) reacts with solid carbon dioxide (dry ice) and is subsequently hydrolyzed with dilute acid, the final product is:

- A) Acetaldehyde
- B) Acetone
- C) Acetic acid
- D) Ethanol

94. Considering the relative reactivity of acid derivatives, which of the following sequences represents the correct decreasing order of reactivity towards nucleophilic acyl substitution?

- A) Acid chloride > Acid anhydride > Ester > Amide
- B) Amide > Ester > Acid anhydride > Acid chloride
- C) Acid anhydride > Acid chloride > Amide > Ester
- D) Ester > Acid chloride > Acid anhydride > Amide

95. In the volumetric estimation of chloride ions by Mohr's method, the pH of the solution is maintained between 6.5 and 9.0. What is the primary reason for avoiding highly acidic conditions ($pH < 6.5$)?

- A) The indicator forms an unreactive complex with chloride ions.
- B) The chromate indicator gets converted into dichromate ions, delaying the endpoint.
- C) Silver chloride becomes highly soluble in acidic media.
- D) Silver precipitates as silver hydroxide before reacting with the chloride ions.

96. Which of the following statements about Thin Layer Chromatography (TLC) is strictly incorrect?

- A) Silica gel is commonly used as a stationary phase.
- B) The R_f value is dependent on the nature of the solvent used.
- C) The R_f value of a specific compound is always greater than 1.
- D) It is primarily an example of adsorption chromatography.

97. A 100 mL of a 0.1 M solution of a weak diprotic acid H_2A is titrated with 0.1 M NaOH. The titration curve shows two distinct, sharp inflection points. What does this observation imply about the acid's dissociation constants, K_{a1} and K_{a2} ?

- A) K_{a1} and K_{a2} are both very small and nearly equal.
- B) K_{a1} is significantly larger than K_{a2} (typically by a factor of 10^4 or more).
- C) K_{a2} is greater than K_{a1} .
- D) The acid acts as a strong acid for both dissociations.

98. Portland cement contains several clinker phases. Which of the following compounds is primarily responsible for the early strength development (within the first few days) of the cement upon hydration?

- A) Tricalcium aluminate (C_3A)
- B) Tricalcium silicate (C_3S)
- C) Dicalcium silicate (C_2S)
- D) Tetracalcium aluminoferrite (C_4AF)

99. Novolac is a linear polymer formed by the condensation of phenol and formaldehyde. Under what conditions is Novolac typically synthesized, and what occurs when it is subsequently heated with hexamethylenetetramine?

- A) Basic conditions; it depolymerizes back to phenol and formaldehyde monomers.
- B) Acidic conditions; it undergoes intensive cross-linking to form Bakelite.
- C) Acidic conditions; it undergoes chain-elongation to form a thermoplastic polyester.
- D) Basic conditions; it reacts to form a straight-chain synthetic rubber.

100. A common pharmaceutical drug serves as both an analgesic and an antipyretic by inhibiting the synthesis of prostaglandins. However, it undergoes hydrolysis in the stomach to yield a product that famously causes gastric irritation. Identify the drug.

- A) Paracetamol B) Aspirin (Acetylsalicylic acid) C) Chloramphenicol D) Ibuprofen

BOTANY

101. Which of the following are the fundamental building blocks of proteins?

- A) Monosaccharides B) Nucleotides C) Amino acids D) Fatty acids

102. The plant cell wall is primarily composed of which structural carbohydrate?

- A) Glycogen B) Chitin C) Cellulose D) Starch

103. Cyanobacteria, which are photosynthetic prokaryotes, are more commonly referred to as:

- A) Red algae B) Blue-green algae C) Green algae D) Brown algae

104. What is the correct scientific botanical name for the medicinally important plant Neem?

- A) Azadirachta indica B) Ocimum sanctum C) Zingiber officinale D) Rauwolfia serpentina

105. A mutant strain of a microorganism utilizes light energy to generate ATP but lacks the ability to fix atmospheric CO₂, strictly requiring external organic compounds as its carbon source. This advanced nutritional mode is best classified as:

- A) Photoautotrophic B) Chemoheterotrophic C) Photoheterotrophic D) Chemoautotrophic

106. An agronomist formulates an herbicide designed to specifically target dicot weeds that possess a cruciform corolla, tetradynamous stamens, and a fruit modifying into a silique. Which weed family is strictly being targeted?

- A) Fabaceae B) Liliaceae C) Brassicaceae D) Solanaceae

107. The majority of the fungal partners (mycobionts) involved in the symbiotic association of lichens belong to which class of fungi?

- A) Basidiomycetes B) Ascomycetes C) Phycomycetes D) Deuteromycetes

108. In an ecologically polluted industrial zone with exceptionally high atmospheric SO₂ emissions, a biologist notes the complete disappearance of a specific organism from tree barks. Upon observing a surviving specimen in a clean area, it shows a complex thallus where fungal hyphae completely enclose algal cells. The eliminated organism is a:

- A) Mycorrhizal root B) Foliose lichen C) Parasitic bracket fungus D) Epiphytic bryophyte

109. Which crucial evolutionary advancement first appeared in Pteridophytes, distinguishing them from Bryophytes and enabling better terrestrial adaptation?

- A) Production of seeds B) Development of true flowers
C) Presence of vascular tissues (Xylem and Phloem) D) A dominant, independent gametophyte stage

110. A swollen placenta, an obliquely placed septum, and axile placentation are the defining diagnostic features of the ovary in which economically important plant family?

- A) Brassicaceae B) Solanaceae C) Liliaceae D) Fabaceae

111. The wood of Pinus (a gymnosperm) is commercially valuable for timber. Anatomically, this wood is characterized as:

- A) Manoxylic and homoxylous B) Pycnoxylic and homoxylous
C) Manoxylic and heteroxylous D) Pycnoxylic and heteroxylous

112. In a typical functional pond ecosystem, what serves as the ultimate source of energy driving the entire system?

- A) Phytoplankton B) Dissolved organic matter C) Solar radiation D) Zooplankton

113. Regardless of the specific ecosystem type (pond, forest, or grassland), which ecological pyramid is always strictly upright?

- A) Pyramid of Number B) Pyramid of Biomass C) Pyramid of Energy D) None of the above

114. The protective ozone layer, which shields Earth's surface from biologically damaging ultraviolet (UV) radiation, is concentrated in which region of the atmosphere?

- A) Troposphere B) Stratosphere C) Mesosphere D) Thermosphere

115. During primary ecological succession occurring on a completely barren rock surface (Xerosere), which specific organisms typically constitute the pioneer community?

- A) Mosses B) Annual grasses C) Foliose lichens D) Crustose lichens

116. A researcher treats a plant cell with a chemical inhibitor that specifically blocks the functioning of the signal recognition particle (SRP), completely preventing actively translating ribosomes from docking to the endoplasmic reticulum. Which cellular outcome will be directly and most immediately impaired?

- A) Synthesis of essential cytosolic enzymes B) Synthesis of lysosomal and secretory proteins
C) Smooth ER lipid and steroid biosynthesis D) Free ribosome assembly in the nucleolus

117. During which specific phase of the eukaryotic cell cycle does DNA replication (synthesis) take place?

- A) G₁ phase B) S phase C) G₂ phase D) M phase

118. If dividing root tip cells of an onion (2n=16) are exposed to colchicine, at which specific stage does the cell cycle primarily arrest, and what genetic condition results if the cell escapes the block without undergoing cytokinesis?

- A) Prophase; Aneuploidy B) Metaphase; Polyploidy C) Anaphase; Polyploidy D) Telophase; Haploidy

119. Which of the following represents the correct chronological sequence of the substages in Prophase I of meiosis?

- A) Leptotene, Pachytene, Zygotene, Diplotene, Diakinesis B) Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis C) Zygotene, Leptotene, Diplotene, Pachytene, Diakinesis D) Pachytene, Leptotene, Zygotene, Diakinesis, Diplotene

120. In higher plants, the Golgi apparatus functions in secretion and cell wall synthesis, but it exists as unconnected, distinct cisternal units scattered in the cytoplasm. These units are called:

- A) Plasmodesmata B) Dictyosomes C) Glyoxysomes D) Sphaerosomes

121. The fundamental physical and functional unit of heredity, consisting of a specific sequence of DNA, is called a:

- A) Chromosome B) Nucleus C) Gene D) Chromatid

122. The universal "start codon" on mRNA, which initiates protein translation and specifically codes for the amino acid Methionine, is:

- A) UAA B) UAG C) UGA D) AUG

123. Which of the following is an exception to Law of Independent Assortment?

- A) Co dominance B) Incomplete Dominance C) Linkage D) Both a and b

124. In a typical Mendelian dihybrid cross involving two parents heterozygous for both traits (AaBb x AaBb), what is the expected phenotypic ratio in the F₂ generation?

- A) 3:1 B) 1:2:1 C) 9:3:3:1 D) 1:1:1:1

125. The genetic phenomenon where a single gene controls or influences multiple, seemingly unrelated phenotypic traits in an organism is known as:

- A) Polygenic inheritance B) Epistasis C) Pleiotropy D) Co-dominance

126. A phenotypically normal man marries a woman whose father was an albino (autosomal recessive) and severely colorblind (X-linked recessive). The woman herself possesses normal pigmentation and normal vision. Assuming the man is heterozygous for albinism, what is the exact probability that their first son will be born both colorblind and an albino?

- A) 1/8 B) 1/4 C) 1/2 D) 1/16

127. Which complex permanent tissue is structurally adapted and responsible for the upward conduction of water and dissolved minerals from roots to leaves?

- A) Phloem B) Parenchyma C) Xylem D) Collenchyma

128. When examining the transverse section of a typical dicot stem under a microscope, how are the vascular bundles arranged?

- A) Scattered randomly throughout the ground tissue B) Arranged in a broken zigzag pattern
C) Arranged radially in a complete, distinct ring D) Grouped exclusively in the center as a solid core

129. Casparian strips are band-like water-impermeable thickenings found on the radial and inner tangential walls of endodermal cells in roots. They are primarily composed of:

- A) Cutin B) Suberin C) Lignin D) Pectin

130. Cell A has a Solute Potential (Ψ_s) of -10 bars and a Turgor Pressure (Ψ_p) of 4 bars. It is placed in direct contact with Cell B, which has a Solute Potential of -12 bars and a Wall Pressure of 5 bars. Based on thermodynamic principles of water relations, what will be the direction of net water movement?

- A) From Cell A to Cell B B) From Cell B to Cell A
C) No net movement will occur D) Water will leave both cells to the environment

131. During an experiment, a plant leaf is treated with a chemical agent that specifically punctures the thylakoid membrane, making it highly permeable to protons (H^+). Based on the chemiosmotic hypothesis, what will be the most immediate and direct consequence during the light-dependent reactions of photosynthesis?

- A) Halting of photosystem II water splitting B) Complete cessation of ATP synthesis
C) Prevention of $NADP^+$ reduction to NADPH D) Dramatic increase in cyclic photophosphorylation

132. Which plant growth regulator is unique because it exists as a gas at standard temperature and pressure, and plays a predominant role in fruit ripening?

- A) Auxin B) Gibberellin C) Cytokinin D) Ethylene

133. Photorespiration is a highly wasteful metabolic process occurring in C₃ plants. Which environmental conditions heavily induce and favor this process?

- A) High CO₂ and low O₂ concentrations B) High O₂ and low CO₂ concentrations, along with high temperatures C) Low light intensity and low temperatures
D) High humidity and optimal CO₂ levels

134. In the Hatch-Slack (C₄) pathway, carbon dioxide is initially fixed in the mesophyll cells to prevent photorespiration. The primary CO₂ acceptor in this initial reaction is:

- A) Ribulose-1,5-bisphosphate (RuBP) B) 3-Phosphoglyceric acid (3-PGA)
C) Phosphoenolpyruvate (PEP) D) Oxaloacetic acid (OAA)

135. A horticulturist aims to commercially propagate a rare medicinal orchid rapidly via plant tissue culture. To specifically induce multiple shoot formation from an undifferentiated callus, which hormonal ratio must be precisely manipulated within the artificial nutrient medium?

- A) High Auxin to Low Cytokinin ratio B) High Cytokinin to Low Auxin ratio
C) Equal parts of Gibberellic Acid and Abscissic acid D) High Ethylene to Low Auxin ratio

136. During fertilization in angiosperms, the actual fusion of a male gamete with the female gamete (egg cell) to form a diploid zygote is technically referred to as:

- A) Triple fusion B) Syngamy C) Plasmogamy D) Karyogamy

137. In certain angiosperms, the primary endosperm nucleus undergoes rapid and repeated mitotic divisions without immediate cell wall formation, creating a liquid multinucleate state (as seen in coconut water). This developmental type is called:

A) Cellular endosperm B) Helobial endosperm C) Nuclear endosperm D) Ruminant endosperm

138. The broad application of scientific techniques to modify and utilize living organisms, cells, or their components to produce agriculturally or medically valuable products for humanity is defined as:

A) Eugenics B) Biopiracy C) Biotechnology / Bio-engineering D) Bio-informatics

139. Which of the following leguminous plants is commonly cultivated and deliberately ploughed back into the soil while still green to enrich it with organic nitrogen and humus?

A) Crotalaria juncea (Sunn hemp) B) Oryza sativa (Rice)
C) Triticum aestivum (Wheat) D) Zea mays (Maize)

40. The excessive discharge of fertilizers into water bodies is called

A) Pollution B) Biological oxygen Demand C) Biological amplification D) Eutrophication

ZOOLOGY

141. In human evolution, which structural milestone is correctly paired with the hominid ancestor that first exhibited it on a regular, habitual basis?

A) Regular production of stone tools - Australopithecus afarensis
B) Habitual, obligate bipedalism with erect posture - Australopithecus ramidus
C) Controlled, domestic use of fire for cooking and protection Homo erectus
D) Intentional ritualistic burial of the dead with grave offerings Homo habilis

142. Which of the following examples represents an evolutionary 'Atavism' (reappearance of an ancestral trait) rather than a simple vestigial structure?

A) The presence of a non-functional nictitating membrane (plica semilunaris) in the human eye
B) A human infant born with a fully formed, mobile external post-anal tail containing skeletal segments
C) The presence of segmentally arranged vermiform appendix structures in modern humans
D) The splint bones located along the sides of a modern horse's third digit

143. In evolutionary genetics, the 'Sewall Wright Effect' refers to which of the following evolutionary forces that can dramatically shift allele frequencies?

A) Gene flow occurring between two highly stable adjacent populations
B) Random genetic drift acting exclusively within small, isolated populations
C) Strong directional natural selection resulting from intense interspecific competition
D) Balanced polymorphism maintained by heterozygous advantage over generations

144. In Phylum Cnidaria, the phenomenon of 'Metagenesis' as classically exhibited by Obelia is best described as:

A) The true alternation of a haploid generation with a diploid generation
B) An asexual polypoid phase alternating with a sexually reproducing medusoid phase
C) The regular modification of larval tissues during retrogressive metamorphosis
D) The behavioral shift from a planktonic free-swimming larva to a parasitic adult

145. Why do nematodes (Phylum Nematoda/Aschelminthes) exhibit a distinct, highly characteristic sinusoidal thrashing movement instead of coordinated peristaltic locomotion?

A) They entirely lack an absolute hydrostatic skeleton
B) Their body wall possesses strictly longitudinal muscles with no circular muscles
C) Their nervous system lacks any longitudinal nerve cords
D) Their outer non-living cuticle is completely inflexible and rigid

146. While most classes of the Phylum Mollusca possess a chitinous rasping tongue-like structure called a radula for feeding, which specific class lacks a radula entirely due to its sedentary filter-feeding lifestyle?

A) Gastropoda B) Cephalopoda C) Bivalvia (Pelecypoda) D) Polyplacophora

147. In protochordate evolution, the dramatic process of 'Retrogressive Metamorphosis' seen in urochordates (such as Herdmania) is characterized by:

A) A highly advanced, motile larva degenerating into a simpler, sessile adult that loses its notochord
B) A simple primitive larva developing complex sensory organs and rapid predatory wings as an adult
C) An adult reverting fully to an embryonic blastula state during adverse winter conditions
D) The transformation of a bilateral organism into a secondary radially symmetrical organism

148. Which type of specialized cellular junction contains transmembrane proteins called connexins that assemble to form a cylinder, directly linking the cytoplasm of adjacent cells for rapid ion transport?

A) Tight junctions (Zonula occludens) B) Desmosomes (Macula adherens)
C) Gap junctions (Nexus) D) Adherens junctions

149. In the histology of loose connective tissue, which specific cell types are directly derived from specialized differentiated B-lymphocytes and actively manufacture circulating antibodies?

A) Fibroblasts B) Plasma cells C) Mast cells D) Macrophages

150. In the skeletal system, which specific sub-type of cartilage is structurally designated as the strongest, possessing dense parallel bundles of thick collagen fibers to resist compression, and where is it located?

A) Hyaline cartilage - Costal cartilages of ribs B) White fibrous cartilage - Intervertebral discs
C) Elastic cartilage - Pinna of external ear D) Calcified cartilage - Supra-scapula of frog

151. Under resting conditions, the negative intra-axonal membrane potential (-70 mV) of a human neuron is primarily established and maintained because the resting neural membrane is significantly more permeable to:

- A) Influxing Sodium ions (Na^+) than effluxing Potassium ions (K^+)
- B) Effluxing Potassium ions (K^+) than influxing Sodium ions (Na^+)
- C) Effluxing Calcium ions (Ca^{2+}) via active transport gates
- D) Incoming negatively charged large plasma proteins

152. In clinical medicine, the long-term relapse of benign tertian malaria caused by Plasmodium vivax years after initial successful blood-stage clearance is directly due to the reactivation of which dormant stage?

- A) Cryptozoites within the skeletal muscles
- B) Hypnozoites residing inside the parenchymal cells of the liver
- C) Sporozoites hidden inside the endothelial lining of capillaries
- D) Ookinete cysts embedded on the external surface of the stomach wall

153. During the sexual phase of Plasmodium within the mosquito stomach, the process of 'Exflagellation' is a highly specialized reproductive phenomenon observed exclusively during the development of:

- A) Macrogametocytes forming a single female ovum
- B) Microgametocytes dividing rapidly to form motile, slender male microgametes
- C) Sporozoites transforming into motile merozoites
- D) The zygote developing its internal microtubular axial filaments

154. In the reproductive anatomy of the earthworm Pheretima posthuma, if you are looking directly for the ventro-median pore through which mature ova are extruded, in which specific segment must you look?

- A) 14th segment
- B) 18th segment
- C) 13th segment
- D) 19th segment

155. The complex excretory system of the earthworm contains both exonephric and enteronephric nephridia. Which of the following correctly classifies the septal nephridia regarding their structure and destination of discharge?

- A) Exonephric; discharging wastes directly to the outer epidermal surface
- B) Enteronephric; discharging nitrogenous wastes directly into the lumen of the intestine
- C) Exonephric; discharging wastes into the buccal cavity for chemical digestion
- D) Peptonephric; functioning as secondary salivary glands in digestion

156. During the physiological processes of aestivation (summer sleep) and hibernation (winter sleep), the frog Rana tigrina shuts down most active systems and relies entirely on which mode of respiration?

- A) Pulmonary respiration via the lungs
- B) Buccopharyngeal respiration via the vascular lining of the mouth
- C) Cutaneous respiration via the moist, highly vascularized skin
- D) Branchial respiration via hidden internal gill arches

157. In the three-chambered circulatory architecture of the frog, deoxygenated blood returning from the bodily tissues enters the sinus venosus and is pumped into the right atrium. Which specific aortic arch carries blood away from the ventricle to the skin and lungs for gas exchange?

- A) Carotid arch
- B) Systemic arch
- C) Pulmocutaneous arch
- D) Lingual arch

158. In the endocrine control of human digestion, which specialized duodenal hormone is secreted in response to the entry of chyme rich in lipids and peptides, directly triggering contraction of the gallbladder and secretion of enzyme-rich pancreatic fluid?

- A) Gastrin
- B) Secretin
- C) Cholecystokinin (CCK)
- D) Enterocrinin

159. The conversion of inactive pancreatic Trypsinogen into its active protease form, Trypsin, which is the foundational trigger for the entire protein-digestion enzyme cascade in the small intestine, is catalyzed by:

- A) Hydrochloric acid (HCl) in the duodenal lumen
- B) Enterokinase (Enteropeptidase) bound to the intestinal brush-border membrane
- C) Carboxypeptidase acting via a negative feedback loop
- D) Bile salts acting in a weakly alkaline environment

160. Roughly 70% of metabolic carbon dioxide (CO_2) is transported within human blood plasma in the form of soluble bicarbonate ions (HCO_3^-). The rapid, high-velocity conversion of CO_2 and H_2O into carbonic acid inside red blood cells is driven by which zinc-containing enzyme?

- A) Carbonic Anhydrase
- B) Alkaline Phosphatase
- C) Carboxyhemoglobinase
- D) Dehydrogenase

161. In pulmonary clinical diagnostics, how is the 'Functional Residual Capacity' (FRC) of the human lungs defined, and which values must be mathematically added to calculate it?

- A) Volume of air left in lungs after forced maximal expiration; Residual Volume (RV) + Tidal Volume (TV)
- B) Volume of air left in lungs after a normal, quiet, unforced expiration; Expiratory Reserve Volume (ERV) + Residual Volume (RV)
- C) Maximum volume of air a person can expire after a forced inspiration; Inspiratory Reserve Volume (IRV) + TV + ERV
- D) Volume of air inspired normally over and above standard tidal parameters

162. In the complex blood coagulation cascade, what is the exact enzymatic function of the activated clotting factor known as Thrombin?

- A) It acts as a tissue phospholipid to stabilize blood platelets at the site of trauma
- B) It proteolytically cleaves soluble Fibrinogen into monomers that polymerize into insoluble Fibrin strands
- C) It directly converts inactive Prothrombin into active Thrombin in a positive loop
- D) It binds tightly to Vitamin K to block plasminogen activation pathways

163. In the electrical conduction architecture of the human heart, after the action potential passes through the Atrioventricular (AV) node, along which specific path does it travel next to reach the ventricular walls?

- A) Internodal tracts directly to Purkinje fibers

- B) The Bundle of His (AV Bundle) branching into right and left bundle branches
- C) The SA node via retro-grade feedback pathways
- D) The coronary sinus venous valves

164. The Juxtaglomerular Apparatus (JGA) is a critical auto-regulatory structure in the kidney. When there is a drop in glomerular filtration rate (GFR) or a drop in blood pressure, the specialized juxtaglomerular cells directly release which enzyme into circulation?

- A) Erythropoietin B) Renin C) Angiotensin II D) Angiotensin-Converting Enzyme (ACE)

165. During the physiological reflex of micturition (urination), which autonomic muscular coordination pattern must take place to allow urine to leave the bladder cavity?

- A) Sympathetic stimulation contracts the detrusor muscle; parasympathetic stimulation contracts the internal sphincter
- B) Parasympathetic stimulation contracts the smooth detrusor muscle; somatic and parasympathetic paths relax the sphincters
- C) Somatic motor paths contract the detrusor muscle; sympathetic signals open the external sphincter
- D) Complete cessation of all pelvic nerve traffic to cause passive elastic collapse of the bladder wall

166. Why is a monosynaptic reflex arc, such as the classic patellar tendon 'knee-jerk' reflex, significantly faster than a polysynaptic withdrawal reflex?

- A) It completely bypasses the spinal cord and travels directly to the cerebral cortex
- B) There is a direct synapse between one afferent (sensory) neuron and one efferent (motor) neuron with zero interneurons involved
- C) It utilizes non-myelinated C-fibers which propagate action potentials via continuous conduction
- D) The sensory receptors involved operate via steroid hormone pathways rather than ion gates

167. Cerebrospinal Fluid (CSF) provides vital physical buoyancy and chemical protection to the central nervous system. Which specialized vascular structures actively secrete CSF, and where are they located?

- A) Arachnoid villi located in the superior sagittal sinus
- B) Choroid Plexuses located within the ventricles of the brain
- C) Pacchionian bodies located in the subarachnoid space
- D) Microglial nests located inside the central canal of the spinal cord

168. When a person shifts their gaze from a distant mountain peak to read small print on a pill bottle right in front of them, which precise structural adjustments take place within the eye during accommodation?

- A) The ciliary muscle relaxes, suspensory ligaments tighten, and the crystalline lens flattens
- B) The ciliary muscle contracts, suspensory zonules slacken, and the elastic crystalline lens bulges to become more spherical
- C) The radial pupillary dilator muscles contract to flatten the curvature of the cornea
- D) The vitreous humor increases in hydrostatic volume to push the retina backward

169. In the physiological maintenance of equilibrium, which specialized vestibular structures are dedicated to sensing dynamic equilibrium (rotational acceleration and angular velocity)?

- A) Maculae containing otoliths located inside the utricle and saccule
- B) Cristae ampullares located within the ampullae of the semicircular canals
- C) Hair cells resting purely upon the tectorial membrane of the cochlear duct
- D) Pacinian corpuscles located within the tympanic cavity

170. Parathyroid Hormone (PTH) acts as the principal regulator of plasma calcium (Ca^{2+}) levels. Which of the following pathways describes its direct mechanism of action?

- A) Inhibits osteoclasts and promotes the deposition of calcium into the bone matrix
- B) Stimulates osteoclastic bone resorption and increases calcium reabsorption in the renal distal tubules
- C) Directly increases the active transport absorption of calcium in the stomach mucosa without vitamin intermediates
- D) Accelerates the excretion of calcium ions in the urine via the proximal convoluted tubules

171. Hormones bind to specific receptors based on their chemical structure. Lipophilic, lipid soluble hormones like Cortisol and Progesterone act by:

- A) Binding to G-protein coupled receptors on the plasma membrane to activate cAMP
- B) Crossing the cell membrane and binding to intracellular or nuclear receptors to alter gene transcription
- C) Opening ligand-gated ion channels on the cell surface
- D) Activating tyrosine kinase cascades at the cell surface membrane

172. During the human female menstrual cycle, what specific endocrine event triggers the onset of menstruation (menses) if fertilization and implantation do not occur?

- A) A sudden surge in Luteinizing Hormone (LH) from the adenohypophysis
- B) The degeneration of the corpus luteum, causing a sharp drop in progesterone and estrogen levels
- C) A massive spike in Human Chorionic Gonadotropin (hCG) secreted by the blastocyst
- D) The complete cessation of Follicle Stimulating Hormone (FSH) production

173. At a molecular level, the profound watery diarrhea ('rice-water stools') characteristic of severe Asiatic Cholera is induced by the cholera toxin permanently activating which intracellular pathways?

- A) Inhibiting the Na^+/K^+ pump, leading to cytolytic bursting of enterocytes
- B) Permanently activating Gs protein, causing overproduction of intracellular Cyclic AMP (cAMP) and massive Cl^- and water secretion
- C) Directly blocking all aquaporin water channels across the descending large intestine mucosa
- D) Degrading the tight junctions between mucosal epithelial cells via protease activity

174. In adaptive cell-mediated immunity, which of the following statements correctly differentiates the functional presentation pathways of MHC Class I and MHC Class II molecules?

- A) MHC Class I molecules present exogenous antigens to CD⁴⁺ Helper T-cells
- B) MHC Class I molecules are on all nucleated cells and present endogenous antigens to CD⁸⁺ Cytotoxic T-cells; MHC Class II molecules are on professional APCs and present exogenous antigens to CD⁴⁺ Helper T-cells
- C) MHC Class II molecules are exclusively restricted to mature red blood cells and platelets
- D) Both MHC classes present identical short carbohydrate antigens to non-differentiated naive B-cells

175. In immunological biotechnology, the production of highly uniform, monoclonal antibodies via Hybridoma technology requires the artificial somatic cell fusion of short lived, activated B-lymphocytes with:

- A) Immortal, highly proliferative cancerous Myeloma cells
- B) Primary macrophages taken from a syngeneic splenic harvest
- C) Attenuated retroviral particles containing reverse transcriptase genes
- D) Totipotent embryonic stem cells grown on feeder layers

176. Why are toxoid vaccines, such as those clinically deployed against Diphtheria and Tetanus, considered uniquely safe and effective immunizing agents?

- A) They consist of live, heavily mutated viruses that can only replicate inside superficial skin layers
- B) They contain purified, chemically inactivated bacterial exotoxins that have lost toxicity but retain full antigenic immunogenicity
- C) They utilize naked plasmid DNA sequences that integrate directly into host autosomal chromosomes
- D) They trigger immediate passive immunity by introducing pre-formed equine globulins

177. In assisted reproductive technology (ART), if a laboratory-fertilized human embryo is transferred into the patient's Fallopian tube at an early cleavage stage containing fewer than 8 blastomeres, the clinical procedure is strictly termed:

- A) Intra-Uterine Transfer (IUT)
- B) Zygote Intra-Fallopian Transfer (ZIFT)
- C) Gamete Intra-Fallopian Transfer (GIFT)
- D) Intracytoplasmic Sperm Injection (ICSI)

178. What is the precise molecular mechanism of action of the crystalline insecticidal protoxins (Cry proteins) produced by the bio-control bacterium *Bacillus thuringiensis*?

- A) They act as neurotoxins to competitively block acetylcholinesterase gates in the insect brain
- B) They are solubilized and activated by the highly alkaline pH of the insect midgut, binding to epithelial cells and creating pores that cause osmotic lysis
- C) They chemically dissolve the external chitinous exoskeleton, leading to rapid fatal dehydration
- D) They block cellular respiration by binding tightly to cytochrome c oxidase systems

179. In animal behavior, how is a 'Kinesis' fundamentally differentiated from a true 'Taxis' behavioral response?

- A) Kinesis is an learned behavior; taxis is a completely unconditioned innate reflex
- B) Kinesis is a non-directional change in the rate of movement or turning frequency in response to a stimulus intensity; taxis is a directed orientation movement toward or away from a stimulus
- C) Kinesis requires complex visual sensory pathways; taxis operates via chemical gradients only
- D) There is no biological difference; both terms describe identical stereotypic pathways

180. The international environmental treaty known as the Ramsar Convention, which Nepal signed into action with the designation of the Koshi Tappu Wildlife Reserve, is globally dedicated to the protection of:

- A) Alpine montane forest ranges containing vulnerable megafauna species
- B) Wetlands of international importance, particularly as waterfowl habitats
- C) High-altitude glacial lake basins threatened by industrial chemical runoff
- D) Migratory avian flyways across landlocked sovereign nations

MAT

181. Pointing to a photograph, a man said, "I have no brother or sister but that man's father is my father's son." Whose photograph was it?

- A) His nephew's
- B) His father's
- C) His own
- D) His son's

182. A man walks 5 km toward South and then turns to the right. After walking 3 km, he turns to the left and walks 5 km. Now in which direction is he from the starting place?

- A) West
- B) South
- C) South-West
- D) North-East

183. If 1st October falls on a Sunday, then 1st November will be...

- A) Monday
- B) Tuesday
- C) Wednesday
- D) Thursday

184. If in a certain code, ROAST is coded as PQYUR, how will SLOPPY be coded in that language?

- A) QNMRNA
- B) QNNRNA
- C) MNMRNA
- D) QNMMNA

185. Moon: Satellite, Earth : ?

- A) Sun
- B) Planet
- C) Solar System
- D) Asteroid

186. The average of the first 50 natural numbers is:

- A) 25.30
- B) 25.5
- C) 25.00
- D) 12.25

187. If the cost price of 12 pens is equal to the selling price of 8 pens, the gain percent is:

- A) 25%
- B) 33.33%
- C) 50%
- D) 66.66%

188. A and B can do a piece of work in 12 days, B and C in 15 days, C and A in 20 days. How long would A take separately to do the same work?

A) 24 days

B) 30 days

C) 40 days

D) 60 days

189. In how many different ways can the letters of the word 'OPTICAL' be arranged so that the vowels always come together?

A) 120

B) 720

C) 4320

D) 2160

190. What is 20% of 30% of 400?

A) 12

B) 24

C) 36

D) 48

191. Statements: All cats are dogs. All dogs are pens. Conclusion: I. All cats are pens. II. All pens are cats.

A) Only Conclusion I follows

B) Only Conclusion II follows

C) Both I and II follow

D) Neither I nor II follows

192. Statement: "Use our product to improve memory." - An advertisement. Assumptions: I. People want to improve memory. II. People pay attention to advertisements.

A) Only Assumption I is implicit

B) Only Assumption II is implicit

C) Both I and II are implicit

D) Neither I nor II is implicit

193. Statements: I. The government has increased the prices of petrol and diesel by 10%. II. The prices of essential commodities have gone up.

A) Statement I is the cause and II is the effect

B) Statement II is the cause and I is the effect

C) Both are independent causes

D) Both are effects of independent causes

194. Statement: A large number of students from a local school failed the final exam. Courses of Action: I. The school should be closed down. II. Teachers should be replaced immediately. III. The causes of failure should be investigated.

A) Only I and II follow

B) Only II and III follow

C) Only III follows

D) All follow

195. Statement: Should exams be banned? Arguments: I. Yes, they cause undue stress to students. II. No, they are necessary to evaluate learning and progress.

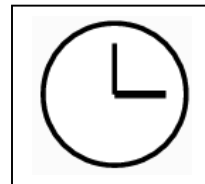
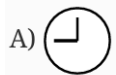
A) Only Argument I is strong

B) Only Argument II is strong

C) Both I and II are strong

D) Neither I nor II is strong

196. Find the mirror image of the clock shown below:



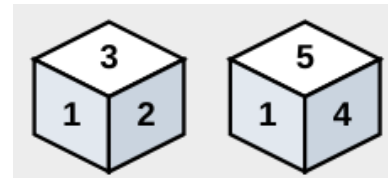
197. Two positions of a dice are shown. What number is on the face opposite to 1?

A) 2

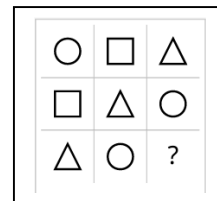
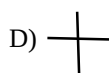
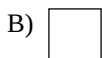
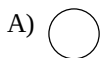
B) 3

C) 5

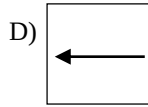
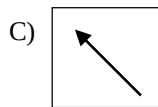
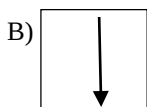
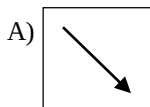
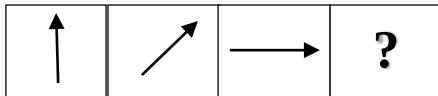
D) 6



198. Complete the figure matrix:



199. Which image correctly completes the series?



200. A square transparent sheet of paper is folded in half from left to right, then in half from top to bottom. A single circular hole is punched exactly in the center of the folded square. When completely unfolded, how will the paper look?

